

⇒ Potential → Electrochemical potential

i.e; potential due to imbalance in electric ~~ions~~ ^{atoms}, chemical one.

→ Electric force $\Rightarrow$ depends on Gradient on Electric field.

→ chemical diffusion force → depends on conc. grad.
-ent

⇒ How much ion on the membrane hold → is said by the Nernst equation usage.

→ In sleeping state → Neurons firing is minimal
In waking state → " more

$\Rightarrow 343 \text{ m/sec} \rightarrow \text{general speed of sound.}$

AP $\rightarrow$ subsonic $\rightarrow$ 10 to 100 m/sec

⇒ myofilaments

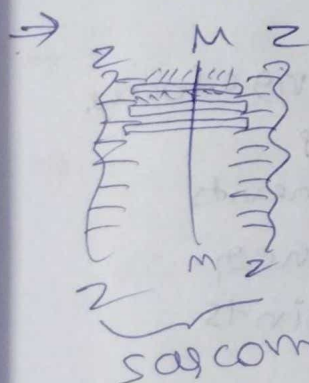
myofibril

~~no~~ 3a7 comele

Thick / Thin ^{myo} filament

M-line $\rightarrow$ associated to thick Myofilament.

Z-line $\rightarrow$ associated to thin myofilament



 → Thick
 → Thin

due to myosin, actin
interplay, ~~and~~

muscle contraction/
expansion occurs

and it is also due

spikes → secrete chemicals

i.e; due to Bond Breakage and formation
b/w spikes and Actin.



Tyndall → connects muscle with bone

⇒ Receptor:- rods, cones in eyes → detects light

⇒ eye anatomy:- sclera → protects eye
choroid → vascular layer
contains all blood vessels

Retinal wall / Retina → has photopigments house

Bulged part of sclera → cornea → where light incident
first refractive medium

Space after cornea → called aqueous humor
Pupil → refracts the incident light

Iris → Aperture work is done by the
→ gives the colour to the eye

Gelly Lens / transparent lens → shape is adaptive

→ ~~shape is~~ focus of the lens ~~can~~ can be changed

↓
ciliary muscles help to change focus of lens

↓
This happens due to ligaments

Ligaments → binds one bone to another through joints / at joints

→ stability is given by this (generally) →

→ In eye → it helps in holding in position → suspensory action & done

→ fovea centralis → Brightest part of eye
All Blood vessels, nerves meet at optic disc
(can) Blind spot

At this point there are no
photo receptors / photo segments
i.e. optic disc passes information
through fibres through
optic nerve and sent that
information to brain.

→ chicken neck → narrow, crucial region
↳ In eye → It is called to
optic nerve ~~part~~
region.

→ afferent ^{means} → sends the information
to brain.

efferent → sends the information
from brain to out.

→ drawing eye figure

→ first Interface to enter light → cornea.
second " " → aqueous
humour

⇒ $ll = rI = \frac{c}{v}$ third → lens
due to change in velocity
of light in different
medium ll develops fourth → then cavity
and then
retina

⇒ $ll = \frac{\sin \theta_1}{\sin \theta_2}$ → By Snell's law

⇒ Crystalline lens → taken on average
because with age + transparency of lens
will be varied and it is varied from
person to person.

wavelengths values → In EM spectrum

Image on retinal wall is inverted due to pin hole
principle.

convexity has to be enhanced when focusing on near object.

→ Far sightedness → Hypermetropia → can see farthest distance
↓
use convex lens

Near sightedness → Myopia → can see nearest objects.
↓
use concave lens

→ Emmetropic vision →

Binocular vision → overlapping region is the region where both field of view of both eyes intersect.

Human → has very good binocular vision

Horses → has very good monocular vision than binocular vision.

Binocular vision is more sophisticated than monocular vision.

→ Eye requires fluid because it keeps the eyeball round and distended by maintaining pressure.

→ viscosity high → fluidity less

→ Aqueous humor → fluidity high

→ Ocular tonometry → due to

→ Glaucoma → Building pressure uniformly and drainage is blocked then this disease occurs.

drainage → releasing fluid that generated regularly in the eye to maintain

- in a certain pressure in eye, not to ↑ the pressure more and bulge eye.

→ due to that disease, optic nerve disk chokes, nutrition supply stops i.e. some fibres can't be sent inside it so that it dies. i.e. signal does not pass and black spot forms there.



~~Ciliary~~
ciliary muscles

↳ that ciliary cells
which produces
water at the
rate of 2.3 lly/min

Schelem's canal → there is some deposition
of water in it
trabecular mesh work → water goes to
through this
and goes to
Aqueous part
and through sweat, the water coming
will go out.

- ⇒ due to deposition, mesh work may be
blocked and pressure rises inside it
due to ~~water~~ more water present
in it.
- ⇒ The pressure will be first generated
in the aqueous humour side and
then transferred to the back
due to Pascal's law.
- ⇒ 10-20 mm of Hg → normal range of
Intraocular pressure
more than that called borderline
high pressure.
there may be < 10 also.
- ⇒ If pressure exceeds 20 — then there is
a problem in the drainage system.
- ⇒ If pressure exceeds 30 — then ~~we~~ we
may lose ~~out~~ our eyes within
short interval of time.

→ Glycoma

→ optic nerve part is made affected by the pressure

→ due to high pressure then there is no cytoplasm flow from eye to brain through optic nerve so that nutrition will not be flow so that often sometime they will die. All will not die at once, so where the fibres will be damaged / lost that region will be as black spot there.

→ Glycoma - It is a progressive disease. Irreversible and it is asymptomatic.

~~→ Cataract - Lens becomes cloudy~~
→ Cataract - Lens becomes cloudy
Replace the lens with an artificial lens.
↳ Transparencancy of lens is gone.

→ Artificial lens → made up of certain polymers (or) biomaterial.

material which can be able to place inside the body.
→ photoreceptors:- Rods, cones

distribution of rods, cone → is not uniform
Brightest part of eye → has cones more than rods

Rods → don't have colour receptors
so we can't see any colour in dim light.
In dim light → rods come into play.

→ cones → gives sense of vision only in the brightest region.

Blue
Green
Red } pigments → primary colours

all colours can be the some mixture of these primary colours

→ zooming the retinal wall;

Ganglion cell → first neuron where signal starts its journey

Bipolar neuron → has 2 modes → on/off depending on the amt of glutamate the Bipolar neuron is set to on/off position.

→ Image forms on the retinal wall (area of the retinal wall), ~~and~~ there are many photoreceptors and send ^{much} information to the brain and then brain ~~the~~ integrates the information and ~~it~~ tell us about the Image.

→ chromophore → also called ~~photo~~ photopsin

→ Night Blindness → due to vitamin A deficiency
(see defn in google)

→ i.e.) vitamin A plays an important role in vision site.

→ Blind spot :- + □

light coming from square ~~at~~ at one point of time will ~~not~~ reach the Blind spot

so that Image can't formed

Blind spot → place where photoreceptors will not be present

⇒ Visual cycle

opsin + retinal

→ ~~photo~~ 1 functional photopigment



The photopigment which has the capacity to absorb light

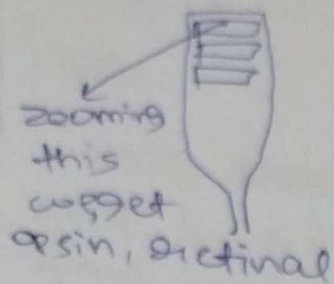
β-carotene gives



source of colour to colour.

of vitamin A

→ In photopigment, retinal is the part which absorbs light



Blue colour pig → Rhodopsin

→

squiggly - loosely

containing photopigment diagram.

→ visual cycle also called

In darkness - Rhodopsin has potential to absorb light

→ Glutamate - inhibitory neurotransmitter

→ Glutamate at synaptic junction.

→ In darkness

(opsin + cis-retinal)
present in dark stage

→ If surrounding is dark, then Na^+ , Ca^{2+} channels will be closed, so Glutamate secretion drastically decreases

so signal passes and we are able to see the surroundings

→ If surrounding is bright light, then Na^+ , Ca^{2+} channels open, so Glutamate secretion increases more, so signal can't pass and we can't see the surroundings.

→ Bleaching → opsin loses its colour (retinal)

→ visual cycle also called as visual cycle of vitamin A (or) visual cycle of retinal.

→ Yellow part rxns → rxns inside opsin

Blue/green part rxns → " outside "

→ cis → trans means that 11-cis retinaldehyde converts into all-trans retinol

RPE 65 → enzyme

→ rods are not important

→ retinal wall looks in orange colour by doc
cones → (rod) pigmental layer
rods → depends on this pigment layer for retinal.

cones → don't depend on this pigment layer for retinal, it has its own.

→ All photopigments can't undergo isomerisation / bleaching at a time.

→ In bright light, lots of photons enter our eye, so lots of rods bleached at a time and lots of cones bleached at a time. So, what matters is which regenerate fastest?

cones will quickly regenerate and visual cycle is going on.

In light adaptation → visual cycle will be going on due to cones.

→ In dark adaptation, if light goes suddenly off, then we see flashes of light due to ^{complete} photopigmentation occurs suddenly we see ~~very~~ very few photons enter our eye.

chance of photon falling into rhodopsin (rods) highest than photopsin (cones)

so rods make the centre stage.

rods take time and regenerate and we see surroundings.

⇒ Dark adaptation (DA) takes more time than light adaptation (LA)

⇒ LA → cones central state → fastest
DA → rods " " → slowest

⇒ rods take more time to regenerate than cones

⇒ rate at which we can perceive sound is more faster than rate at which we can perceive light.

⇒ Ears has some receptors which sense body equilibrium.

⇒ Inner ear → sophisticated part

cartilage → soft bone

⇒ we have semivoluntary muscles which connects cartilage to brain (inside ear)

⇒ outer ear ends at eardrum / tympanic membrane

⇒ Beyond eardrum, we have 3 connected bones (smallest bones in body)

→ ~~outer part to channel~~

⇒ Outer Ear :-

starts with auricle → auditory canal → eardrum

Inferior → line at bottom

superior → line at top

Anterior →

posterior →

eardrum also called as tympanic membrane

⇒ ceruminous glands secrete cerumen due to because, it is a cleansing function. it solidifies the dust and comes down slowly.

perforated eardrum → hole inside eardrum
↓
due to which vibration won't be perceived so that hearing will be effected.

↓
The treatment is generally not required as it healed it automatically but ~~the~~ ~~per~~ in that period we have to protect ~~out~~ our ~~ear~~ ear by cotton.

→ Impacted cerumen → cerumen will not do function properly and solidify the drum and pain arises
↓
treatment is water Jet inside the ear.

⇒ middle ear → connects outer ear to inner and transfers the ~~info~~ information.

⇒ stapes is directly connected to inner ear via the oval window.

⇒ oval window → Bulging in
Round window → Bulging out

⇒ Hammer, Anvil, stirrup → Three Bones / ossicles inside middle ear

↓
⇒ they are connected using ligaments and the ligaments provides stability to the connection, they give them the spring like property.

- ⇒ If there are no ligaments, then with a vibration the bones dislocate and can't come to the before position
- ⇒ Ligament has damper like properties, which helps the bones to not oscillate and come back to before position slowly
- ⇒ stapedius muscle → smallest muscle in body
- ⇒ It helps to protect the inner ear if the sound exceeds threshold value, it helps to not vibrate the inner ear for that sound.
- ⇒ Vibration is nothing but a sound wave.
- ⇒ At eardrum → pressure $55 \text{ mm}^2 \rightarrow A_e$
At oval window → " $3.2 \text{ mm}^2 \rightarrow A_o$
- ⇒ as $A_e > A_o \Rightarrow P_e < P_o$ ($\because F = PA \Rightarrow P \propto \frac{1}{A}$)
Amount of amp = $\frac{55}{3.2} \approx 17$ times ($\because \frac{A_1}{A_2} = \frac{P_2}{P_1}$)
- ⇒ $f_{malleus} \& malleus = f_{incus} \& incus$
displacement of malleus > incus
 $\Rightarrow f_{incus} > f_{malleus}$
↓
here also we are getting some amount of amplification of 5 times
- ⇒ totally we get 22 times amplification in middle ear, which converting ~~in~~ vibrating while transferring information from outer ear to inner ear via middle ear.
- ⇒ Middle ear will amplify the incoming signal.

→ why ~~that~~ this amplification is required?
even though inner ear is sensitive enough?

Ans: Due to the prop called Inertia (i.e.)
In air → Inertia is less
Inside ear → It has to travel inside liquid, It requires more Inertia.

→ ∴ It requires high pressure to travel inside fluid. so, this amplification is required.

→ sound wave should travel inside in the medium to propagate.

→ labyrinth → maze
means

→ cochlea → has receptors of hearing sensation

→ Inside duct → endolymph present outside and inside duct space → contains perilymph

→ cochlea → 3D spiral.

yellow colour one in cochlea in physiology of hearing → called organ of corti

→ As the fluid flows, the vibration flows throughout the cochlea and certain frequency will be matched with the pitch of organ of corti then it becomes activated and the signal will be passed.

Anatomy of Inner ear → 2 main parts

1) 3b spiral part → cochlea

2) Another → vestibular apparatus

Sensory receptors for hearing heard in cochlea

Body Balance sensors present in vestibular apparatus

⇒ Auditory transduction

↓
happen at cochlea converting into electrical signals

⇒ Stapes Bone connected to, Inner ear at a junction called oval window

⇒ All blue colour part → perilymph (not actually blue, just it's inside ear)

⇒ It has 2 channels. → upper channel → flows inward
lower channel → flows outward

Signal starts at oval window → upper channel (bulge inwardly) ↓ reaches ground (bulge outwardly) ← lower channel ← Apex window ends at

⇒ Signal passage (or) flow of perilymph in upper channel and lower channel

contains perilymph fluid. they are not isolated, they are connected.

⇒ Inflow them, ^{upper and lower channels} there is a cochlea duct filled with endolymph.

⇒ upper and lower channel meet at apex.

⇒ upper channel, cochlea duct separates by a membrane called vestibular membrane
lower channel " " " "

Beside membrane

⇒ Above organ of Corti there is tectorial membrane

- ⇒ Horizontal ducts — Nerves
- ⇒ Vestibular membrane will vibrate first and then endolymph feels vibration and then transfers to tectorial membrane against its hair cells
- ⇒ then hair cells bend and signal passed to nerves and travels in dedicated neural path.
- ⇒ Entire Basilar membrane does not vibrate simultaneously, only to the pitch of the sound matches, then that part vibrates and sends signal to brain by nerve fibres.
- ⇒ Dynamic equilibrium → controls by
- ⇒ Maculae → look about static equilibrium
- ⇒ Hair cell
 - top of ~~hair~~ hair cell
 - it has hair bundle
- ⇒ Semicircular ducts (3)
 - ↓
 - they are arranged in 3 sides
 - (semiliquid gelly like structure)
- ⇒ Otolithic membrane → secreted by support cells.
 - ↓
 - They contain otoleths
- ⇒ when we change the orientation of our body, orientation of the maculae changes so that otoleth moves, then hair cells bend and changes (occurs), so that the fibres inside i.e. vestibular nerve detect signal.
- ⇒ If there is a continuous imbalance, then brain sends many signals, which sense it as ~~more~~ disease i.e. it confused and sends continuously.

→ Ampulla has cista



→ which detects signals
of imbalance

in semicircular duct

→ Three 1/2 rotations are independent to each other

→ movement

→ Auricle - collect information/signals and sends inside

→ eardrum - which senses vibration first

→ Auditory ossicles → transfer from outer to inner ear,
done amplification

→ maintaining pressure b/w outer, inner

→ ear - Eustachian tube.

Revision

→ Neural circuit

→ series circuit is simplest among all neural circuits.

→ neural circuits are complicated than electrical circuits

→ Analysing neural circuits means analyzing the information passing through it

→ plasticity / neuroplasticity → prop of neural and regeneration circuits

→ Learning something means ~~creating~~ firing neurons ~~more~~

and doing that many times, ~~fast~~ neuroplasticity increases.

- ~~the~~ some reptiles has capability of replication of neurons many times but we, the mammals can regenerate limitedly.
- V-matrix of size 18K